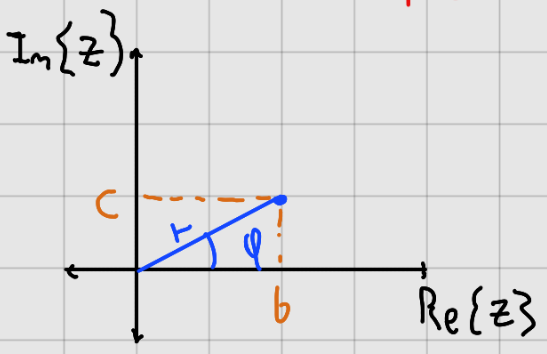
Viernes 6/3 Clase Practico

Prof Agustin Martinez Balsa

Numeros Complejos: (Repaso)



$z = b + jc \rightarrow z = r e^{j\varphi}$ $\quad \cos\varphi = \dfrac{e^{j\varphi} + e^{-j\varphi}}{2}$

$r = \sqrt{b^2 + c^2} \quad \theta = tg^{-1}\left(\dfrac{c}{b}\right)$

$e^{j\varphi} = \cos\varphi + j\,\text{sen}\,\varphi \qquad \text{sen}\,\varphi = \dfrac{e^{j\varphi} - e^{-j\varphi}}{2j}$

Ejemplo:

$Z_1 = a + jb = r_1 e^{j\varphi_1}$

$Z_2 = c + jd = r_2 e^{j\varphi_2}$

$Z_1 + Z_2 = (a+c) + j(b+d)$

$z_1 + z_2 = (a+jb).(c+jd) = r_1 r_2 e^{j(\varphi_1 + \varphi_2)}$

$Z_1^* = a - jb = r_1 e^{-j\varphi_1}$ (Conjugado)

Ejercicios:

1) $Z = (Z_1 . Z_2) + Z_3$

$Z_1 = 2 + j$

$Z_2 = 3 e^{j\frac{2\pi}{4}}$

$Z_3 = 1 - 5j$

$Z_2 = 3j$

$a = 3 \cos 1/2\,\pi$

$b = 3\,\text{sen}\, 1/2\,\pi$

$Z_1 . Z_2 = 6j - 3$

$Z = -2 + j$

$Z_1 = \sqrt{5}\, e^{j\,0{,}4626}$

$Z_1 . Z_2 = 3\sqrt{5}\; e^{j(0{,}4626 + 1/2\,\pi)}$

$= 3\sqrt{5}\, e^{j2{,}03}$

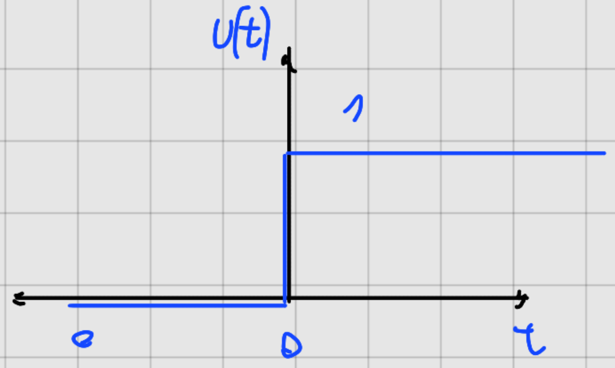
$= 3\sqrt{5} \cos 2{,}03 \quad + j\, 3\sqrt{5}\,\text{sen}\, 2{,}03$

$= -3 + j6$

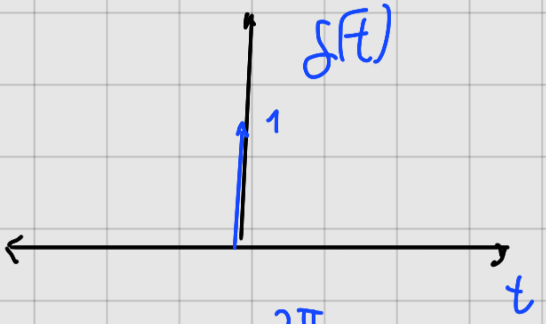
$Z = -2 + j$

Repaso de Señales

$$U(t) = \begin{cases} 0 & t < 0 \\ 1 & t < 1 \end{cases}$$



$\delta(t)$



$$ce^{a} = e^{j\omega_0 t} = e^{j\omega_0 (t+T)} = e^{j\omega_0 t} \cdot \underbrace{e^{j\overbrace{\omega_0 T}^{2\pi}}}_{1} \rightarrow 2\pi = \omega_0 T$$

$$T = \frac{2\pi}{\omega_0}$$

Pasos para ejercicios de transformaciones:

1. Traslado
2. Escalamiento
3. Inversión

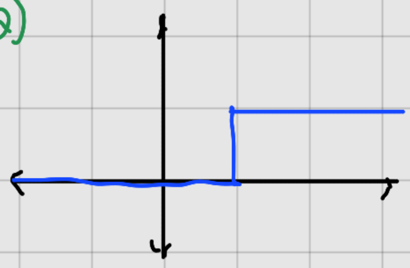
$U(2t-1)$

$2t - 1 = 0$
$t = 1/2$

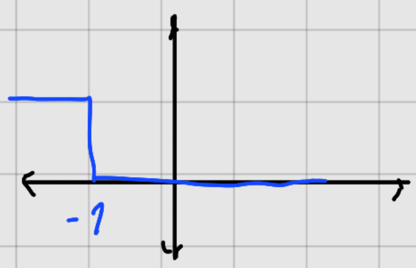
Dibujar:

a) $u(t-1)$
b) $u(-t-1)$
c) $u(-2t-1)$
d) $-3u(t-3)$

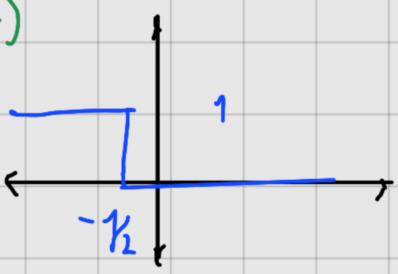
a)



b)



c)



d)

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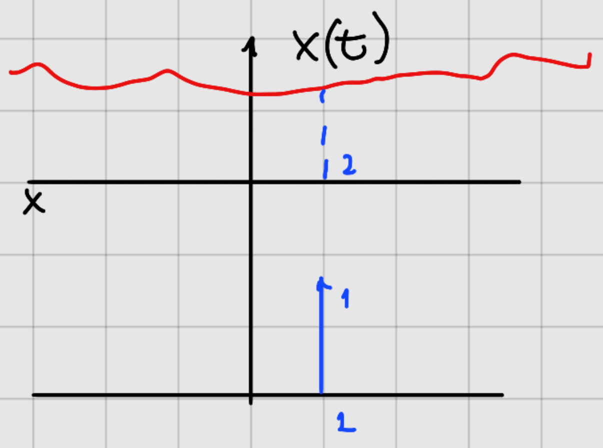
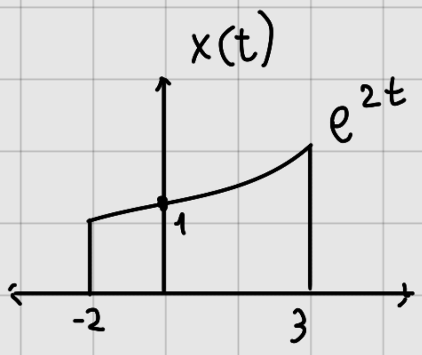
$x(t).\delta(t) = x(0).\delta(t)$ | $x(t).\delta(t-2) = x(2).\delta(t-2)$



$$x(t)\,U(t-t_0) = \begin{cases} 0, & t<t_0 \\ x(t), & t>t_0 \end{cases}$$



$$x(t) = e^{2t}\left[U(t+2) - U(t-3)\right]$$

$$y(t) = x(t).\left[U(t-t_0) - U(t-t_1)\right]$$

$$y(t) = \begin{cases} 0 & t<t_0 \\ x(t) & t_0<t<t_1 \\ 0 & t>t_1 \end{cases}$$

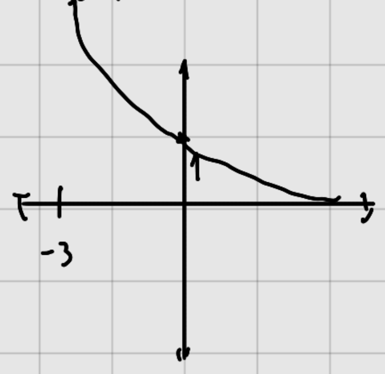
a) $x_1(t) = e^{-2t}\,U(t+3)$

b) $x_2(t) = -\frac{1}{2}\,U(t+4) + \frac{1}{2}\,U(t+2)$

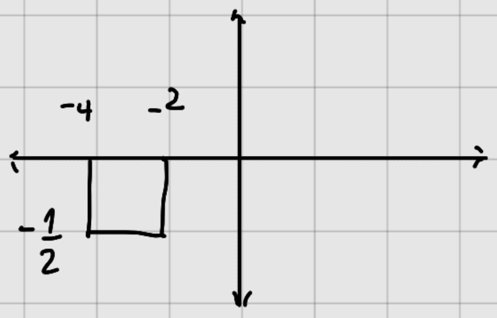
c) $x_3(t) = -\frac{1}{2}\,U(t+4) + \frac{1}{2}\,U(t+2) + 1$

d) $e^{-2t}\,U(t-1) + e^{t}\,u(-t-5)$

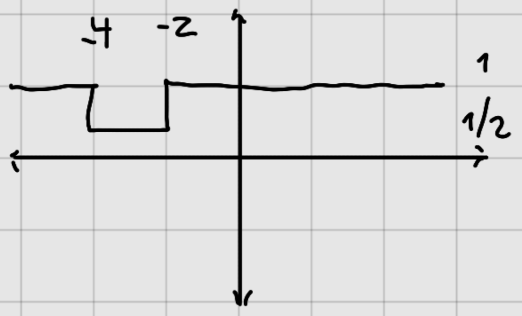
a)



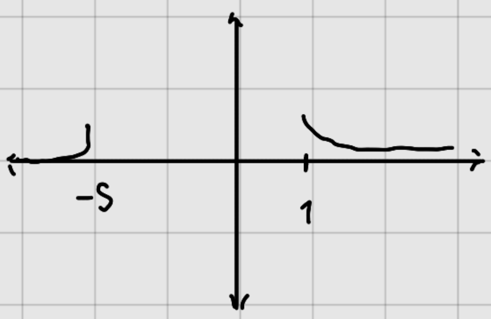
b)



c)



d)



Par $x(t) = x(-t)$

Impar

$x(t) = -x(-t)$

$x(t) = \text{par}\{x(t)\} + \text{impar}\{x(t)\}$

$\text{Par} = \frac{1}{2}\left[x(t) + x(-t)\right]$

$\text{Impar} = \frac{1}{2}\left[x(t) - x(-t)\right]$

La suma de señales periodicas entrega una señal periodica.

$z(t) = \tilde{x}(t) + \tilde{y}(t)$

$z(t) = x(t + k_1 T_1) + y(t + k_2 T_2)$

$T = \text{m.c.m}(T_1, T_2)$

Ejemplo

$X(t) = \cos\left(\frac{2\pi}{3} + \frac{\pi}{4}\right) \to T_1 = \frac{2\pi}{w} = \frac{2\pi}{\frac{2\pi}{3}} = 3$

$Y(t) = \cos\left(\frac{4}{3}\pi + \frac{\pi}{2}\right) \to T_2 = \frac{2\pi}{w} = \frac{2\pi}{\frac{4\pi}{3}} = \frac{3}{2}$

$T = 3 \to \text{mcm}\left(3, \frac{3}{2}\right)$